Visualization of Carbon Portfolios: Structural Bottleneck of Economic Development

Peixuan Wang
pwang441@gatech.edu

Shuo Li
sli3031@gatech.edu

Yinuo Li
yli3865@gatech.edu

Wanrong Gao
wgao301@gatech.edu

Figure 1: Deforestation Caused by Human Arson

Abstract

The world's resources are being exploited and the population is growing. Despite progress in areas like poverty reduction, we need to look beyond just economic metrics in development as every development has consequences. Our goal is to promote sustainable economic growth by paying attention to development by-products, especially carbon emissions which are linked to economic achievements. In our project, we'll visualize carbon emissions and GDP growth in major countries and regions worldwide, and assess the utilization efficiency of carbon emissions. Through interactive features, users can track performance, helping them better understand these trends.

Keywords

Carbon Emission, Economic Growth, GDP, Data Envelopment Analysis, Visualization

1 Introduction

Against the backdrop of continuous development and utilization of global resources and population growth, although positive progress has been made in areas such as poverty reduction, the traditional development measurement system dominated by economic indicators is gradually becoming one-sided. Behind every economic growth comes an undeniable byproduct, especially carbon emissions, which increasingly pose a challenge to global sustainable development. Therefore, development should not only focus on the expansion of economic scale, but also on the environmental costs it brings.

This project aims to promote sustainable economic growth and focuses on the dynamic relationship between carbon emissions and economic development. We propose that carbon emissions should be one of the important evaluation dimensions when assessing development outcomes. Specifically, we visualized the carbon emission levels and GDP growth of major countries and regions around the world, and further analyzed their carbon utilization efficiency (calculated using DEA method) to explore the structural bottlenecks in carbon emission utilization in each country.

The main innovation of this project lies in not only systematically analyzing the economic environmental relationship at the macro level, but also delving into the micro sector level, drawing "carbon portfolios", and using an interactive visual dashboard to help users intuitively understand the interaction trend between carbon emission structure and economic development. We hope that through this project, we can provide data support and decision-making references for formulating more effective sustainable development policies and optimizing resource allocation.

CSE 6242, Final Project Report
© 2025

2 Literature Survey

2.1 Macro perspective

Early research often explored the relationship between economic growth and carbon emission through the Environmental Kuznets Curve (EKC) hypothesis[1], which posits an inverted U-shaped relationship between per capita income and environmental degradation. The theory suggests emissions rise with income initially but decline after a certain threshold due to technological advancements, structural shifts, and policy interventions[2]. While some studies found empirical support for the EKC in specific regions or country groups[3] , its universality, especially for global pollutants like CO_2, is widely debated, with many studies finding different relationship patterns or questioning its applicability outside high-income nations[4].

Relatedly, the concept of decoupling, which is weakening the link between economic growth and environmental pressure, has been extensively studied. Relative decoupling (emissions growing slower than GDP) is common globally, driven by energy efficiency improvements and the rise of clean energy[5]. Absolute decoupling (emissions falling while GDP grows) is observed mainly in advanced economies[6]. Key drivers include technological progress, structural economic shifts towards services, renewable energy deployment, electrification, and policy interventions like carbon pricing. While informative, these macro perspectives do not form the basis of our specific analysis, which focuses on efficiency.

2.2 The Efficiency Perspective

A growing body of research shifts the perspective by treating carbon emissions not just as an externality but as an inherent (undesirable) output or, conceptually, as an input factor within the economic production process[7]. This allows for the application of production theory and efficiency measurement techniques to evaluate how effectively economic systems generate value (GDP) while managing their carbon footprint.

Within this framework, various terms are used, often interchangeably, leading to some conceptual ambiguity. Common metrics include: Carbon Intensity (CO_2/GDP), which is a simple single-factor measure; Carbon Productivity (GDP/CO_2), which is the inverse of carbon intensity, indicating economic output per unit of emission; Carbon Emission Efficiency (CEE), Total Factor Carbon Productivity(TFCP) and Green Total Factor Productivity (GTFP) which are more comprehensive measures, typically derived from frontier analysis methods (like DEA or SFA), that account for multiple inputs (e.g., capital, labor, energy) and outputs (GDP as desirable, CO_2 as undesirable) [7][8]. Our project aligns with this total-factor perspective, specifically using DEA.

Data Envelopment Analysis (DEA) is a non-parametric linear programming technique widely used for assessing the relative efficiency of decision-making units (DMUs) with multiple inputs and outputs. Its key advantages include not requiring a pre-specified functional form and its ability to handle multiple inputs/outputs simultaneously, making it suitable for complex environmental-economic assessments[9].

In the context of carbon efficiency, DEA models are adapted to incorporate CO_2 emissions. Common approaches treat emissions as an undesirable output (to be minimized alongside maximizing desirable outputs like GDP) or directly as an input, reflecting the view that emissions are a necessary 'cost' or factor consumed in production. Models like the Slacks-Based Measure (SBM) or Directional Distance Function (DDF) are specifically designed to handle undesirable outputs.

Our project adopts the perspective of treating sectoral carbon emissions as an input to generate GDP growth as the output. While Stochastic Frontier Analysis (SFA) is an alternative parametric method , DEA's non-parametric nature is often preferred when the underlying production technology is complex or unknown[10].

DEA-based studies have assessed carbon efficiency across various scales. Globally, efficiency trends have fluctuated, with some evidence suggesting improvements in recent decades, potentially linked to technological advancements. Comparisons between country groups like BRICS and G7 often find G7 nations generally exhibit higher resource or sustainable development efficiency, although BRICS nations sometimes show faster improvement rates. Studies also highlight significant efficiency variations within countries and across different industrial sectors. Factors influencing efficiency include technology, industrial structure, energy mix, policy interventions, and economic development levels.

While numerous studies have applied DEA to assess carbon or environmental efficiency at national or regional levels, there is a gap in systematically analyzing and visualizing the sectoral carbon efficiency profiles across a wide range of countries over an extended period (1990-2021). Existing research often presents efficiency scores numerically, which can obscure complex patterns and hinder intuitive understanding, especially for policymakers.

Our project addresses this gap by:

1. Applying DEA specifically using sectoral carbon emissions as inputs and GDP growth as the output for major countries from 1990 to 2021.

2. Developing novel visualization techniques to represent these country-sector efficiency results as "carbon portfolios."

3. Utilizing these visualizations, potentially within an interactive dashboard framework , to facilitate the identification of structural bottlenecks in economic development pathways across different sectors and countries. Geospatial visualization techniques might also be relevant for displaying country-level results.

By integrating efficiency analysis with advanced visualization, this research aims to provide a clearer, more actionable understanding of how different economies utilize carbon emissions for growth across various sectors, thereby supporting the transition towards more sustainable development patterns.

3 Dataset

3.1 Data Source and Description

This project is based on global data provided by Our World in Data. There are four main datasets used in this study include:

- Annual carbon emissions data[11]: time range from 1750 to 2023. The data feature include the total annual carbon dioxide (CO_2) emissions (in billion tons)

- GDP per capita data[12]: time range from 1990 to 2023. The data feature includes GDP per capital (calculated in 2021 international dollars)

- Historical population data[13]: time range of 10,000 BCE to 2023. The data feature includes the total global population (in billions)

- Carbon emissions data by sectors[14]: time range from 1990 to 2021. The data feature includes the carbon emissions of various sectors (buildings, industry, land use change and forestry, other fuel combustion, transport, manufacturing and construction, energy production, electricity and heat, bunker fuels).

All datasets contain the following core features: Entity (country), Code (abbreviation of country name) and Year.

The dataset analyzed in the report covers data from different time spans, some of which can be traced back to 10,000 BCE, providing the possibility of long-term series analysis. All data is based on organized and validated global public data sources, ensuring the reliability and scientific validity of the data.

3.2 Data Processing and Results

In this part we introduce the procedures of processing to get the datasets that we used to analyze on.

Entity	Code	Year	GDP per capita	Annual CO2 emissions	Population (historical)	GDP in Total	Annual CO2 emissions per capita
Afghanistan	AFG	1950	1156	84272	7776182	8989266392	0.010837194911333094

Figure 2: Dataset Structure and Example

First, we merged the carbon emissions, GDP, and population datasets using Entity, Code, and Year as the key attributes. This integration ensures that all relevant indicators are aligned for each country and year. Below is the merged dataset:

We found that there is a significant amount of missing data for some countries before 1950. To ensure data reliability and consistency, we decided to begin our analysis from that point onward. However, as shown in the figure below, starting from 1980 provides the most comprehensive coverage across countries, making it the most appropriate starting point for our analysis.

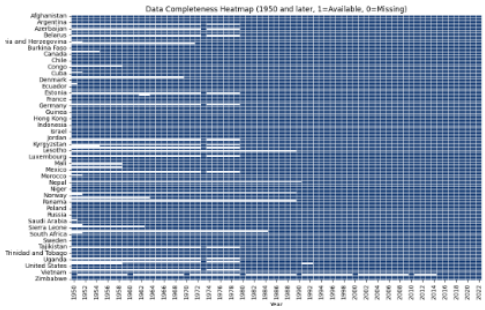

Figure 3: Data Completeness Heatmap

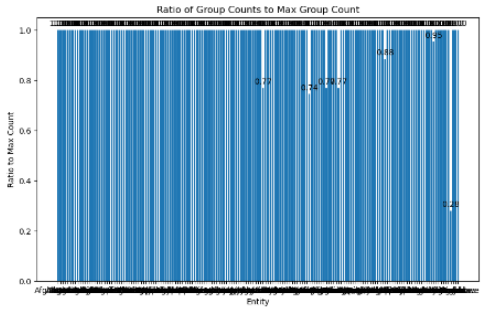

Figure 4: Ratio of Group Counts to Max Count

We calculated the completeness ratio can visualized it in the figure above. It can be said that by truncating our data and keeping those after 1980s, we ensuring the completeness of data without dumping too much information.

4 Methodology

4.1 Visualization Analytics

We began by compiling a rich set of global indicators: total CO_2 emissions, per capita emissions, total GDP, GDP per capita, total population and sector-level CO_2 breakdowns (Buildings; Transport; Energy Production; Manufacturing & Construction; Land Use & Forestry; Other Fuel Combustion; Electricity & Heat; Bunker Fuels)—from trusted source Our World in Data. This integrated dataset underpins our interactive dashboard, ensuring consistency across countries and over the last decades.

We then developed an interactive dashboard featuring a dynamic Choropleth map, a bubble scatter plot (per-capita GDP vs. per-capita emissions), multi-series line charts, bar charts, and a rose chart. By interacting with these visuals, users gain a bird's-eye perspective on the evolving link between economic growth and carbon emissions. They can identify persistent heavy emitters, observe emerging economies climbing the GDP-emissions frontier, and spot early signs of decoupling in mature markets. This macroscopic overview lays the foundation for deeper efficiency and sectoral analyses later in the report.

4.2 Data Envelopment Analysis

We're going to apply the DEA method to calculate out a series of Technical Efficiency (TEs) for each unit and their Weight Vectors (WVs) for each year. DEA functions by constructing a conceptual production frontier, where all combinations of input-output configurations can not be out-performed by any member from existing study objectives. This method has long been used by researchers for evaluations of performance and also providing insights for further improvements based on WVs.

Farrell[15] first proposed the concept of efficiency, dividing production efficiency into technical efficiency and allocative efficiency to measure the production frontier. Subsequently, Charnes, Cooper, and Rhodes[16] developed the CCR model. They[17] further modified the CCR model to the BCC model. Both the CCR and BCC models measure radial efficiency, allowing inputs or outputs to

increase or decrease in equal proportion. We are using the BCC model[17] in our project.

The DEA model constructed is shown as follows.

The most important concept in DEA method is DMU. DMUs are decision-making entities, who can make decisions to decide how to manage their production plans and have these plans' outcomes recorded in dataset. In this study, all countries discussed via this model can be deemed as DMUs.

The core idea is to construct a virtual DMU as a linear combination of potentially all existing DMU from our dataset. The purpose of DMU is to set a model as the most efficient DMU that is theoratically possible to exist. It's possible to achieve because it's merely a combination of existing DMUs, which means it can at least be reached just by tuning the DMU's production scale. Thus, we're going to have the weights of each DMU's own production configurations to show their contributions in the virtual DMU's configuration, the vector of such weights can be denoted as . To find the best possible virtual DMU is essentially a process to solve a Optimization Problem, so the optimization formulation of our DEA model can be represented as

$$\min \quad \theta \tag{1}$$

$$\text{s.t.} \sum_{i=1}^{n} \lambda_i x_{ij} \leq \theta x_{ij} \tag{2}$$

$$\sum_{i=1}^{n} \lambda_i y_{ir} \geq y_{kr} \tag{3}$$

$$\sum_{i=1}^{n} \lambda_i = 1 \tag{4}$$

$$\lambda_i \geq 0 \tag{5}$$

The formula is the classic BCC-model DEA formulation. In it, each country is treated as a DMU, and we solve a linear program that minimizes its efficiency score under the following constraints:

1. Input constraints: $\sum_{i=1}^{n} \lambda_i x_{ij} \leq \theta x_{ij}$ for each input i, ensuring that the weighted sum of other DMUs' inputs does not exceed the target DMU's inputs scaled by θ.

2. Output constraints: $\sum_{i=1}^{n} \lambda_i y_{ir} \geq y_{kr}$ for each output r, requiring that the weighted sum of outputs at least matches the target DMU's output.

3. Variable returns to scale: $\sum_j \lambda_j = 1$, which allows the model to account for scale efficiencies.

4. Non-negativity: $_j \geq 0$ and ≥ 0.

Here, the decision variables $_j$ define the "virtual DMU" as a convex combination of peers, and the optimal θ^* (between 0 and 1) measures how close the target DMU is to the efficient frontier– a score of 1 means it is on the frontier, while a score below 1 indicates inefficiency and room for improvement.

4.3 Case Study

To complement our DEA results, we computed the L_2 distance (Euclidean distance) between each country's normalized nine-sector emission vector and those of all efficient peers; this distance serves as a structural similarity score. For each less-efficient country, we then select the closest efficient country on the efficient frontier as its reference or "role model."

After the pairing, we conducted a case study based on the comparison between the two instances, and recognize the bottleneck of the economic mode of less effective entity, then giving customized suggestions.

5 Analysis

5.1 Macroscopic Analysis

Over the past thirty years, global CO_2 emissions have become increasingly concentrated among a few major economies. In 1990, the top five emitters (United States, Russia, China, Japan and Germany) accounted for over 55% of total emissions. By 2020, rapid industrialization in China and India has shifted this balance: China alone produced nearly 30% of global CO_2, while India rose into the top three. Meanwhile, many advanced economies plateaued or even slightly reduced their absolute emissions, reflecting early efforts at de-carbonization.

Per-capita emissions show a similar widening gap. Small, resource-rich nations such as Qatar, Kuwait, and the United Arab Emirates remain at the top of the per person ranking, while populous developing countries, including Bangladesh and Nigeria, stay below 2 t. Over three decades, average per-capita emissions in emerging markets have nearly tripled, while mature economies saw only modest growth or slight declines.

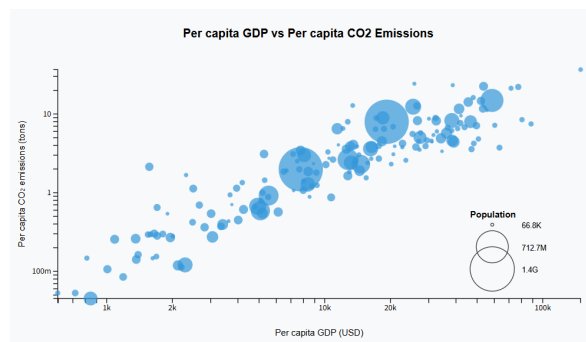

Figure 5: Scatter Plot of Per capita GDP vs Per capita CO_2

The scatter plot of GDP per capita versus CO_2 emissions per capita (Figure 5.2) reveals a remarkably strong log-linear relationship. Beyond this threshold, many high-income countries—such as Germany, France, and Japan—begin to flatten or even reduce their per-person emissions, indicating early stages of decoupling. Countries like China exemplify the typical development trajectory: in 1990, China sat in the lower-left quadrant (low GDP, low emissions), but by 2020 it migrated into the mid-range both economically and environmentally. South Korea and Turkey follow similar upward sweeps, while the United Kingdom, Germany, and France move rightward in GDP but stabilize or drop in emissions.

Finally, a small cohort of countries - China, the United States, India, Russia, and the EU - have consistently dominated the total emissions leader board, collectively responsible for more than two-thirds of global CO_2. In contrast, South Korea, Turkey, and Iran climbed from outside the top ten in 1990 to top-ten emitters by 2020. At the same time, the United Kingdom, Germany and France have

slipped down the rankings, showcasing successful, policy-driven emissions reductions. These long-term patterns underscore both the persistence of major emitters and the dynamic rise and fall of emerging and mature economies.

5.2 Microscopic Analysis

In this section, we present our detailed analytical framework and findings, beginning with a microscopic examination of sectoral emissions profiles, followed by the computation of TE via DEA, and concluding with illustrative case studies of Brazil and Argentina.

We conducted a microscopic analysis by decomposing each country's total carbon dioxide emissions into 9 key sectors: Buildings, Transport, Energy, Other Fuel Combustion, Bunker Fuels, Manufacturing & Construction, Electricity & Heat, and Land Use & Forestry (We only kept 8 in our dashboard for intrinsic inbalence in data, making sure the pattern is easier to recognize). To ensure comparability, we normalized the sectoral emissions using min–max scaling, producing a nine-dimensional "carbon portfolio" vector for every country-year. These portfolios reveal each nation's unique emissions structure and highlight the industries that dominate its environmental footprint.

After above preprocessing, we applied the Banker–Charnes–Cooper (BCC) DEA model to evaluate each country's TE in transforming carbon emissions (inputs) into economic output (GDP). Treating countries as DMUs, we solved a linear program for each country-year from 1990 to 2021. The resulting TE scores range from 0 to 1, with 1 indicating that a DMU lies on the efficient frontier. To validate our results, we performed robustness checks, including the exclusion of outliers and sensitivity analyses on input aggregation.

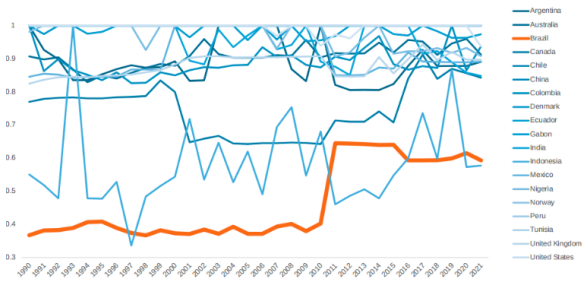

Figure 6: TE trends

After calculating the TEs, we illustrated the trend of each country's efficiency in the polyline chart above. We noticed that Brazil, whose trend is highlighted with orange color, keeps ranking low for most of the time, which triggered our interest. According to our pairing principle, the potential model for Brazil to follow is Argentina. We are going to do case study based on that in the following paragraphs.

5.3 Brazil & Argentina: Development bottleneck and potential solution

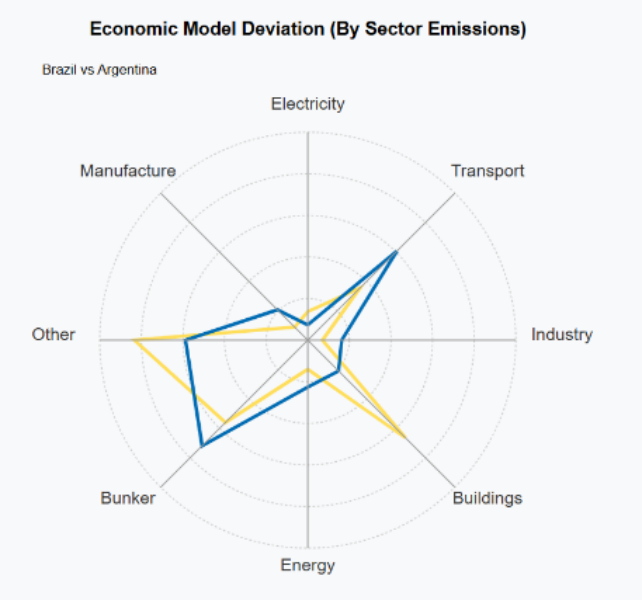

Figure 7: Radar Chart of CO_2 Emission Portfolios

In the provided rose chart, the blue represents the distribution of industrial carbon emissions in Brazil, while the yellow represents the industrial carbon emission structure of Argentina located on the efficient frontier.

The rose chart (Figure 6) contrasts Brazil's sectoral carbon portfolio (blue) with Argentina's (yellow). Brazil's emissions have long been dominated by Land Use & Forestry, which alone accounts for over 40 % of its total CO_2 output. This reflects an economic model built on extensive deforestation and slash-and-burn agriculture: raw timber and cleared land are exported after minimal processing, and the newly cleared acreage is largely devoted to soybean cultivation—much of which is destined for overseas markets. Consequently, Brazil's TEs languished between 0.35 and 0.45 throughout the 1990s, only rising to about 0.65 following the 2003 PPCDAm initiative and its enforcement through 2016. Despite this improvement, Brazil remains well below the efficient frontier.

In contrast, Argentina consistently achieves locating on efficient frontier most of the time by maintaining a balanced emissions structure across agriculture and energy, supported by stable enforcement of environmental regulations. Its diversified sector mix serves as a template for sustainable growth, demonstrating that economic development and emissions control can advance in tandem.

Customized Suggestions: To emulate Argentina's success, Brazil should move up the global value chain by expanding domestic processing and manufacturing—rather than relying solely on exporting raw commodities—and strengthen long-term legislative safeguards against short-term resource exploitation. By enacting stricter forest protection laws, incentivizing investment in renewable energy and agro-industry, and aligning development policies with social equity goals, Brazil can foster a resilient economic environment that

delivers higher returns domestically while safeguarding its forests for future generations.

6 Visualization

This project focuses on global per capita CO_2 emissions and GDP, utilizing multiple interconnected visualizations, dynamic time evolution, and feature switching to comprehensively explore the relationships between carbon emissions, economic development, sectoral distributions, historical trends, and technical efficiencies.

Various forms of visualization are employed, including a world map, scatter plot, time series trend plot, bar charts, and radar charts. Combined with interactive design, this enhances the readability and exploratory capability of the data.

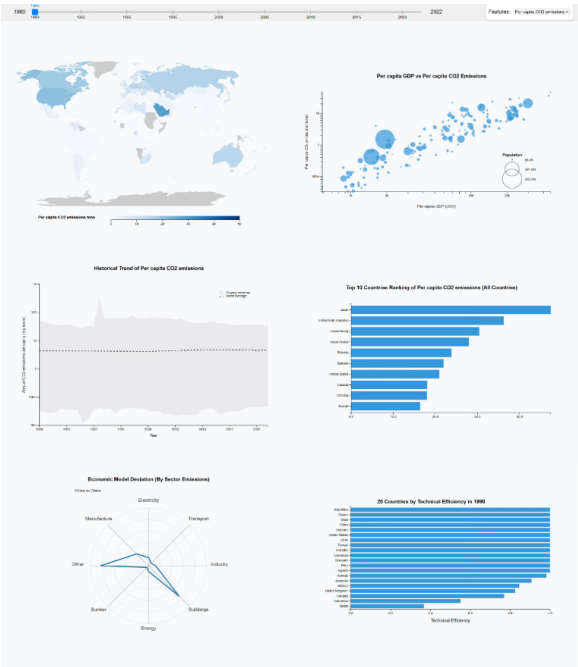

Figure 8: Visualization Dashboard

6.1 Interactive Features

6.1.1 Timeline Interaction. A timeline is placed at the top of the page, allowing users to freely scroll between 1980 and 2022.

When the timeline changes, the following visualizations automatically update: World Map, Scatter Plot, Top 10 Countries by CO_2 Emissions Bar Chart, Sectoral Emissions Radar Chart, Technical Efficiency Bar Chart.

This enables users to observe the evolution of various indicators over time intuitively.

6.1.2 Features Selection Filter. A filter is provided in the top-right corner, allowing users to select different features for display, including:Per capita CO_2 emissions, Per capita GDP, CO_2 emissions (total), GDP (total), Population.

After selecting a feature, the data and visualizations for the map, scatter plot, and other related charts are updated accordingly, supporting exploration of the relationships between global economy and carbon emissions from multiple perspectives.

6.1.3 Map Click Interaction. When clicking on a country in the world map, the corresponding country will be highlighted in orange across other charts, such as the scatter plot, top 10 ranking bar chart, and TE line chart.

This allows users to focus on a specific country and track its performance across different dimensions.

6.2 Chart Descriptions and Analysis

1. World Map

This world map displays the per capita CO_2 emissions of countries globally for the year 2010, using a color gradient to represent emission levels.

Taking Brazil as an example, the data popup shows its population (193.7M), GDP ($2.8T), and total CO_2 emissions (440.3M tons).

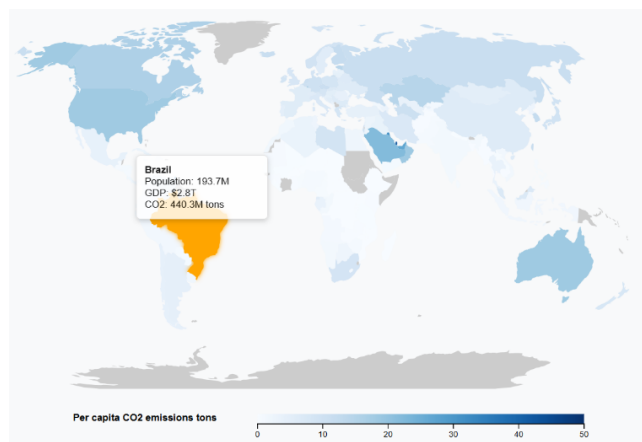

Figure 9: Heatmap of Per capita CO_2 emissions(ton)

From the map, we observe that North America, Australia, and the Middle East generally have darker colors, indicating higher per capita emissions. Meanwhile, central Africa and parts of South Asia show extremely low emissions, highlighting the unequal global distribution of carbon emissions.

2. Scatter Plot

This plot shows the relationship between per capita GDP and per capita CO_2 emissions for each country, with bubble size representing population.

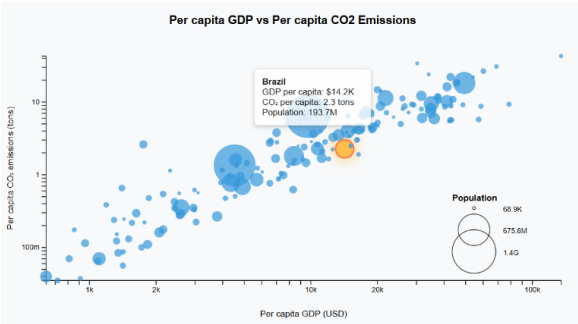

Figure 10: Scatter Plot of per capita GDP and per capita CO_2 Emissions

There is a clear overall positive correlation: countries with higher per capita GDP tend to have higher CO_2 emissions.

However, there are outliers, such as some Middle Eastern countries with moderate per capita GDP but extremely high emissions, likely due to their oil-based economies.

Countries with large populations (e.g., China and India) are represented by particularly large bubbles, reflecting their global impact.

3. Time Series Trend Graph

This graph shows the trend of Brazil's per capita CO_2 emissions from 1980 to 2022, compared against the global average. The gray shaded area represents the range between the minimum and maximum values across all countries.

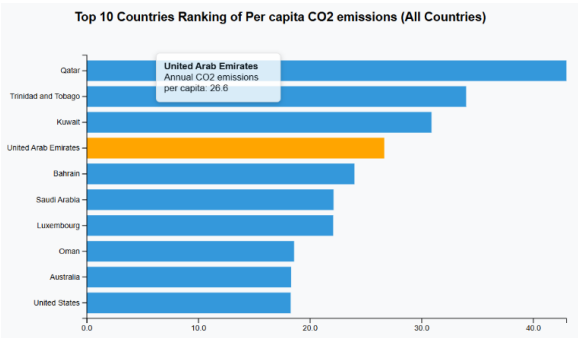

Figure 11: Time Series Trend Graph

Brazil's per capita emissions have remained relatively stable with a slight upward trend, consistently staying below the world average.

4. Top 10 Countries Ranking Bar Chart

This chart lists the top 10 countries with the highest per capita CO_2 emissions.

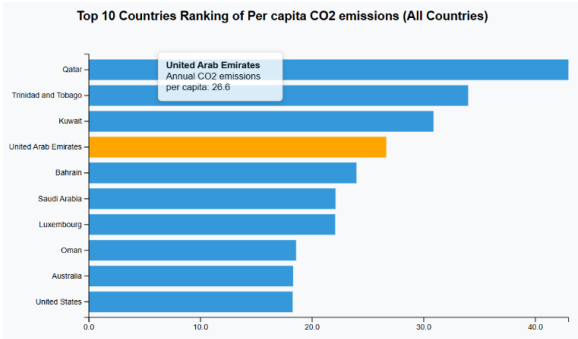

Figure 12: Top 10 Countries Ranking Bar Chart

Qatar stands out with per capita emissions approaching 45 tons. The top 10 are mainly small oil-exporting or energy-dependent countries, such as the United Arab Emirates, Kuwait, Bahrain, and Saudi Arabia. This reflects that resource-based economies disproportionately contribute to global carbon emissions.

5.Radar Chart

This chart compares the target country (Brazil) and the reference country (Argentina) in terms of carbon emissions across different sectors (Electricity, Transport, Industry, Buildings, Energy, Bunker, Other, Manufacture).

The radar chart shows the proportional contribution of each sector, where the closer the shapes, the more similar the sectoral structures.

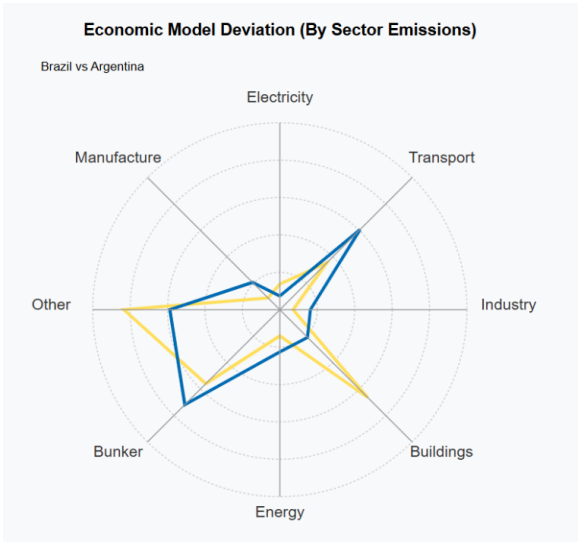

Figure 13: Radar Chart of CO_2 Emission Portfolios

Argentina (yellow) and Brazil (blue) exhibit high similarity across most sectors, especially in Transport, Industry, and Manufacture. However, in the Bunker (fuel consumption) and Energy (energy production) sectors, Brazil's emissions are significantly higher than Argentina's.

Brazil can learn from Argentina's experience in energy usage efficiency and fuel consumption management. Particularly in the energy and fuel sectors, Argentina's lower emissions provide a potential path for Brazil to optimize its energy structure and reduce overall carbon emissions, contributing to sustainable development.

6. Technical Efficiency Bar Char

This chart displays the technical efficiency rankings for 20 countries in the year 2010.

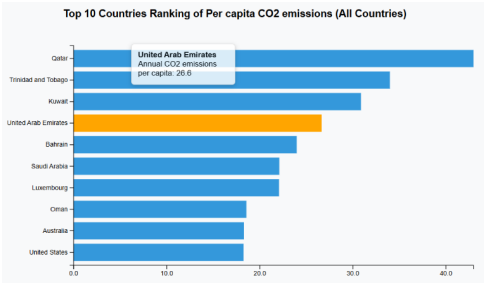

Figure 14: Technical Efficiency Bar Char

Chile, Denmark, and Mexico are among the top performers with near-perfect TE scores. Brazil ranks lowest, suggesting that compared to other countries, Brazil achieves lower output for the same level of resource input or suffers from higher resource wastage.

7 Conclusion and Discussion

7.1 Summary of Findings

In this project, we systematically explored the relationship between carbon emissions and economic growth through both macroscopic and microscopic perspectives. By integrating a variety of datasets and employing interactive visualization methods alongside DEA, we were able to capture dynamic global trends, sectoral structures, and technical efficiencies from 1990 to 2021.

7.2 Key Insights

At the macroscopic level, we observed that global CO_2 emissions have become increasingly concentrated among a few major economies over the past three decades, and a strong positive relationship persists between per capita GDP and per capita emissions. Meanwhile, relative and absolute decoupling trends have emerged in some advanced economies, indicating potential pathways toward sustainable growth.

At the microscopic level, sectoral analysis revealed distinctive carbon emission structures across countries. Using DEA, we quantified the TE of converting carbon emissions into economic output. Our case study comparing Brazil and Argentina demonstrated how structural dependencies—such as Brazil's reliance on land-use change and forestry—can hinder efficiency improvements. Argentina's diversified and balanced emission profile offered valuable insights for policy reform and strategic adjustment.

7.3 Limitations

However, some limitations remain. DEA results are sensitive to the choice of input and output variables, and the model does not account for embodied carbon emissions in supply chains. Moreover, improving carbon efficiency alone does not guarantee overall environmental or social sustainability.

7.4 Future Directions

In future research, we recommend expanding the analysis to include broader environmental indicators, integrating embodied emissions accounting, and conducting more granular sectoral or regional studies. By combining efficiency analysis with richer sustainability metrics, more comprehensive and actionable insights can be generated for policymakers and stakeholders.

This project provides a novel integration of DEA-based carbon efficiency analysis and interactive visualization. It offers actionable insights into the structural bottlenecks of economic development across countries, supporting more targeted and sustainable policymaking.

References

[1] Tong Zhang, Wanru Zheng, Hui Li, Weiming Chen, and Hua Liao. Economic development and co2 emissions in china: a multi-dimensional analysis. *Journal of the Asia Pacific Economy*, pages 1–22, 2025.
[2] Claudien Habimana Simbi, Fengmei Yao, and Jiahua Zhang. Sustainable development in africa: A comprehensive analysis of gdp, co2 emissions, and socioeconomic factors. *Sustainability*, 17(2):679, 2025.
[3] Wang Jie and Khan Rabnawaz. Renewable energy and co2 emissions in developing and developed nations: a panel estimate approach. *Frontiers in Environmental Science*, 12:1405001, 2024.
[4] Ming-Chung Chang, Jin-Li Hu, and Heng-Chu Chang. Resource efficiency and productivity changes in the g7 and brics nations. *Polish Journal of Environmental Studies*, 27(6), 2018.
[5] Yali Li and Jianhua Xiao. Environmental efficiency assessment of the us pulp and paper industry using an sbm-dea model. *BioResources*, 15(4):7796, 2020.
[6] Fu Hua, Majed Alharthi, Weihua Yin, Muhammad Saeed, Ishtiaq Ahmad, and Syed Ahtsham Ali. Carbon emissions and socioeconomic drivers of climate change: empirical evidence from the logarithmic mean divisia index (lmdi) base model for china. *Sustainability*, 14(4):2214, 2022.
[7] Shuangjie Li, Wei Wang, Hongyu Diao, and Liming Wang. Measuring the efficiency of energy and carbon emissions: A review of definitions, models, and input-output variables. *Energies*, 15(3):962, 2022.
[8] Minyoung Yang and Jinsoo Kim. A critical review of the definition and estimation of carbon efficiency. *Sustainability*, 14(16):10123, 2022.
[9] Jing Zhu. Analysis of carbon emission efficiency based on dea model. *Journal of Discrete Mathematical Sciences and Cryptography*, 21(2):405–409, 2018.
[10] George Katharakis, Maria Katharaki, and Theofanis Katostaras. An empirical study of comparing dea and sfa methods to measure hospital units' efficiency. *International Journal of Operational Research*, 21(3):341–364, 2014.
[11] Robbie M. Andrew and Glen P. Peters. The global carbon project's fossil co emissions dataset (2024v17), 2024.
[12] World Bank. Gdp per capita (current us$), 2024. Retrieved from Our World in Data.
[13] United Nations, Department of Economic and Social Affairs, Population Division. World population prospects 2024, online edition, 2024. Retrieved from Our World in Data.
[14] Our World in Data. Per capita co emissions by sector, world, 2021, 2021.
[15] Michael James Farrell. The measurement of productive efficiency. *Journal of the royal statistical society series a: statistics in society*, 120(3):253–281, 1957.
[16] Abraham Charnes, William W Cooper, and Edwardo Rhodes. Measuring the efficiency of decision making units. *European journal of operational research*, 2(6):429–444, 1978.
[17] A Charnes and William W Cooper. The non-archimedean ccr ratio for efficiency analysis: A rejoinder to boyd and färe. *European Journal of Operational Research*, 15(3):333–334, 1984.